

HW# 6
Deep Learning (DL_525.733-8VL)
Due date: April 2nd, 2026

- Design a conditional GAN (cGAN) architecture using the code in reference [1] and MNIST dataset to perform the following mappings (i.e., odd digit to even digit image image):

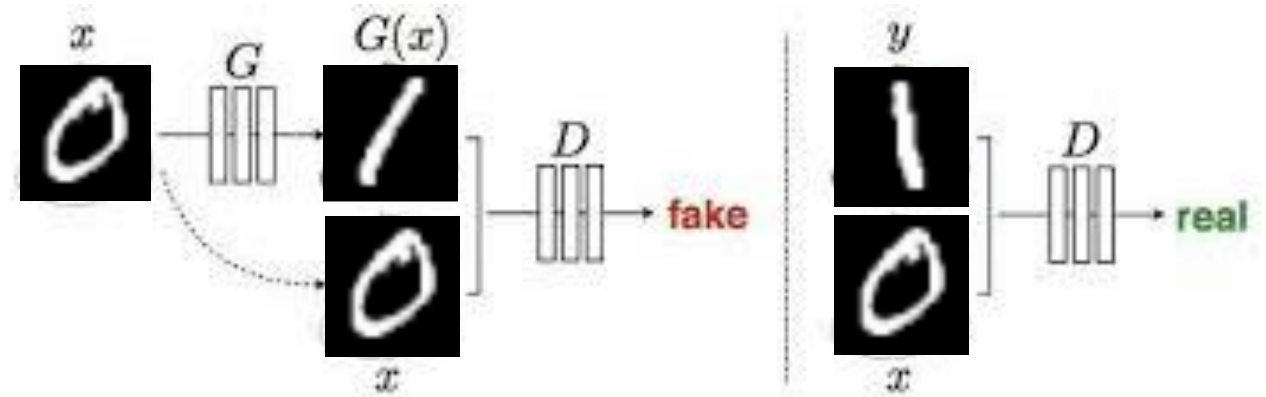
Input digit image  $\Rightarrow$ output image 

Input digit image 2 $\Rightarrow$ output image 3

Input digit image 4 $\Rightarrow$ output image 5

Input digit image 6 $\Rightarrow$ output image 7

Input digit image 8 $\Rightarrow$ output image 9



- Show your training curves for G and D. Show several examples for each digit for the output of the generator.**

[1] (Conditional GAN) “Image-to-Image Translation with Conditional Adversarial Networks,” Phillip Isola Jun-Yan Zhu Tinghui Zhou Alexei A. Efros Berkeley AI Research (BAIR) Laboratory, UC Berkeley